Practice Midterm

Last Name: _____

First Name: _____

Student ID: _____

Signature: _____

Instructions: This practice midterm is a bit longer than the real midterm, so that you have more problems to practice on.

Please do not write below this line.

Question	Points	Score
1	20	
2	15	
3	10	
4	15	
5	18	
6	0	
7	12	
Total:	90	

1. Consider the tensor $\mathcal{A} \in \mathbb{R}^{2\times 2\times 3}$ and the matrix $B \in \mathbb{R}^{2\times 2}$ where:

$$\mathcal{A}(:,:,1) = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \quad \mathcal{A}(:,:,2) = \begin{bmatrix} -4 & 0 \\ 3 & 1 \end{bmatrix} \quad \mathcal{A}(:,:,3) = \begin{bmatrix} 2 & -7 \\ 1 & 5 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 5 \\ -2 & 1 \end{bmatrix}$$

   (a) (5 points) Can we compute $\mathcal{A} \times_1 B$? If yes, compute this. If no, explain why not.

   (b) (5 points) Can we compute $\mathcal{A} \times_3 B$? If yes, compute this. If no, explain why not.

   (c) (5 points) Explain, in words, what it means to unfold a tensor.

   (d) (5 points) Given, in words, one application in data science where tensors might be used.

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2. Consider the matrix $A \in \mathbb{R}^{3\times 4}$, given below together with its singular value decomposition:

$$A = \begin{bmatrix} 0.5377 & 0.8622 & -0.4336 & 2.7694 \\ 1.8339 & 0.3188 & 0.3426 & -1.3499 \\ -2.2588 & -1.3077 & 3.5784 & 3.0349 \end{bmatrix}$$

$$= \begin{bmatrix} 0.2137 & -0.9463 & -0.2427 \\ -0.2814 & 0.1782 & -0.9429 \\ 0.9355 & 0.2698 & -0.2282 \end{bmatrix} \begin{bmatrix} 5.622 & 0 & 0 \\ 0 & 2.8458 & 0 \\ 0 & 0 & 1.7117 \end{bmatrix} \begin{bmatrix} -0.4438 & -0.2781 & -0.7852 & -0.3303 \\ -0.1992 & -0.3907 & -0.1235 & 0.8902 \\ 0.5575 & 0.5049 & -0.6044 & 0.2626 \\ 0.6727 & -0.7177 & -0.0537 & -0.1719 \end{bmatrix}$$

(a) (5 points) Give the definition of the matrix 2-norm, $\|A\|_2$.

(b) (5 points) Using the information given, compute $\|A\|_2$.

(c) Compute the matrix ∞-norm, $\|A\|_\infty$.

(d) (5 points) Compute, using the information given, the best rank 2 approximation to A .

3. Consider the least squares problem: $\min_{\mathbf{x}\in\mathbb{R}^n} \|A\,\mathbf{x}-\mathbf{b}\,\|_2$ where $A \in \mathbb{R}^{m\times n}$.

   (a) (6 points) What is the QR decomposition of A? How would one use this to solve this least squares problem?

   (b) (4 points) Recall that $\kappa(A)$ denotes the condition number of A. Which situation would be preferable: $\kappa(A) = 5$ or $\kappa(A) = 10^4$? Justify your answer.

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4. For the following statements, state whether they are true or false. If false, explain what is wrong with the statement or provide a corrected version.

   (a) (5 points) For every matrix $A \in \mathbb{R}^{n \times n}$ there exists an upper triangular matrix U and a lower triangular matrix L with ones on its diagonal such that $A = LU$.

   (b) (5 points) For every matrix $B \in \mathbb{R}^{m \times n}$ there exists an orthogonal matrix $Q \in \mathbb{R}^{m \times m}$ and a matrix $R \in \mathbb{R}^{m \times n}$ with $R = \begin{bmatrix} R_1 \\ 0 \end{bmatrix}$ where $R_1 \in \mathbb{R}^{n \times n}$ is upper triangular such that $B = QR$.

   (c) (5 points) For every non-negative matrix $X \in \mathbb{R}^{m \times n}$ and every $r < \min\{m, n\}$ there exists non-negative matrices $H \in \mathbb{R}^{m \times r}$ and $W \in \mathbb{R}^{r \times n}$ such that $X = WH$.

5. (a) (8 points) Write pseudocode for the k-means algorithm.

(b) (5 points) Draw one data set, consisting of two clusters, for which k-means will correctly find these clusters.

(c) (5 points) Now, draw one data set, again consisting of two clusters, for which k-means **will not** find these clusters.

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6. Consider the following text processing problem. Suppose that we have four documents, and we select 5 key words. The matrix X, given below, shows the number of times each keyword appears in each document. For example, $X_{23} = 1$ means that keyword 2 appears once in document 3.

$$X = \begin{bmatrix} 3 & 0 & 1 & 2 \\ 5 & 1 & 1 & 4 \\ 0 & 7 & 2 & 1 \\ 2 & 5 & 1 & 0 \\ 1 & 4 & 1 & 4 \end{bmatrix}$$

Use the tf-idf scheme to make a new, weighted, term matrix $\tilde{X}$. You do not need to normalize the columns.

7. (12 points) Consider the linear program:

$$\begin{aligned} \text{minimize } & 2x_1 + x_2 \\ \text{subject to: } & x_1 + x_2 \leq 9 \\ & x_1 \leq 5 \text{ and } x_2 \leq 7 \\ & x_1 \geq 0 \text{ and } x_2 \geq 0 \end{aligned}$$

Sketch the feasible region and find a solution to this problem.